ADHWAIDE KRISHNA

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PROFESSIONAL SUMMARY

AI & Machine Learning graduate specializing in Agentic AI, RAG systems, Knowledge Graphs, Cognitive Memory Architectures, and Applied LLM Engineering. Experienced building multi-agent research platforms, computer vision memory systems, and AI-powered decision-support applications using Python, FastAPI, React, and open-source LLMs.

TECHNICAL SKILLS

AI Systems & LLMs: Ollama, Hugging Face, Google AI Studio, Mistral, Llama, Qwen, Gemma, Local LLM Deployment

AI/ML Techniques: Prompt Engineering, RAG (Retrieval-Augmented Generation), Vector Databases, ChromaDB, Semantic Search, Knowledge Graphs, Agentic AI, Multi-Agent Systems

ML Frameworks: YOLOv8, TensorFlow, PyTorch, NLP, Computer Vision, Object Detection, Speech Recognition, Text-to-Speech

Programming: Python, JavaScript, C++, R, HTML, CSS

Frameworks & APIs: Django, FastAPI, React, REST APIs, WebSockets

Data & Tools: Pandas, NumPy, Power BI, Tableau, Git, GitHub, Google Colab, AWS (Familiar)

PROFESSIONAL EXPERIENCE

Python Developer Intern | Techmindz, Kochi *02 Dec 2024 – 22 Dec 2024*

- Built full-stack Django web application with integrated frontend, backend, authentication, and database optimization
- Developed RESTful API endpoints enabling efficient and secure client-server communication
- Deployed production-ready application with performance tuning, security hardening, and scalable architecture

AI / ML PROJECTS

V1B3 Cognitive Memory System *Apr 2026 – Jun 2026*

Tech: Python, FastAPI, React, SQLite, OpenCV, YOLOv8 |

- Engineered a five-layer cognitive memory architecture combining perception, episodic, semantic, spatial, and procedural memory for persistent object tracking and retrieval.
- Developed a custom **1280-dimensional visual embedding pipeline** and object re-identification system optimized for **CPU-only deployment** without GPU dependencies.
- Built an explainable reasoning dashboard featuring knowledge graph visualization, memory replay, and step-by-step decision tracing for transparent AI decision-making.

CatalystAI – Stock Intelligence & Research Platform *May 2026 – Jun 2026*

Tech: Python, Ollama, NetworkX, SQLite, Knowledge Graphs, Vector Embeddings | GitHub: github.com/Adhwaide/CatalystAI

- Built an AI-driven stock intelligence platform mapping **100+ Indian companies across 15+ sectors** using knowledge graph traversal and causal relationship analysis.
- Engineered a multi-factor scoring engine combining relationship strength, revenue exposure, source reliability, and market validation signals to rank investment opportunities.
- Integrated local LLMs through Ollama to extract entities from macroeconomic events and generate explainable investment theses.

Autonomous Research Analyst (ARA) *Jun 2026 – Jun 2026*

Tech: Python, Ollama, ChromaDB, RAG, Multi-Agent Systems, Knowledge Graphs |

- Developed a multi-agent research platform capable of ingesting and analyzing content from research papers, RSS feeds, and code repositories through autonomous AI workflows.
- Implemented RAG pipelines, semantic search, and knowledge graph construction using local LLMs and vector embeddings for contextual information retrieval.
- Built reasoning agents for hypothesis generation, causal exploration, opportunity discovery, and multi-agent debate-based analysis.

Digital Immortality Platform – AI Persona Simulation System | Lead Developer *Mar 2025 – Present*

Tech: Python, FastAPI, React, Ollama, Vector Databases, Semantic Embeddings |

- Architected a RAG-based digital persona system combining structured identity data with semantic memory retrieval for context-aware interactions.
- Engineered a custom vector-memory architecture supporting episodic, semantic, and behavioral memory layers to maintain personality consistency across conversations.
- Implemented an autonomous memory reflection pipeline that continuously synthesizes and stores new memories from interaction histories.

EDUCATION

B.Sc. in Artificial Intelligence & Machine Learning *May 2026*

CMS College of Science and Commerce, Coimbatore, Tamil Nadu

Relevant Coursework: Data Structures, Python, C++, R Programming, IoT, Business Data Analytics

CERTIFICATIONS & ADDITIONAL

Certifications: Data Science Foundations (Coursera) | Power BI Essentials (Microsoft) | Data Visualisation – TATA Forage

Domain Interests: Agentic AI Systems | Financial Markets & Quantitative Research | Computer Vision | Autonomous Decision Systems

Languages: English (Fluent) | Malayalam (Native) | Hindi (Fluent) | Tamil (Fluent)